Math 421 - Work **Spring 2025**

Exam 1

Name: Solutions

> The exam contains 7 pages (including this cover page) and 6 questions. The total number of possible points is 50. You have 75 minutes to complete the exam.

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **No outside resources allowed**. You may not use any notes, calculators, textbooks, or other resources on this exam. Make sure all electronic devices are silenced and put away.
- **Partial credit will be given**, so write down as much as you know even if some part of the proof might be missing.
- **Read the questions carefully**, make sure you answer the question being asked and understand the statement of the problem.
- **Answer all questions in the space provided**, you may use the back of pages as scrap paper, but only what is written under the question will be graded.

Do not write in the table to the right.

Question	Points	Score
1	11	
2	10	
3	8	
4	7	
5	7	
6	7	
Total:	50	

1. Consider the following 3 subsets of $\mathbb{R}$ and use them to answer the following questions. Write your answers in the boxes provided.

   - $A = \{x \in \mathbb{R} : x < -1 \vee x \geq 4\}$ $= (-\infty, -1) \cup [4, \infty)$
   - $B = \{x \in \mathbb{N} : x \text{ is prime}\} = \{2, 3, 5, 7, 11, \ldots\}$.
   - $C = \{x \in \mathbb{Z} : x^2 < 25 \wedge x \text{ is odd}\}$. $= \{-3, -1, 1, 3\}$

   Determine the following - that is, specify the set in terms of its specific elements, represent it as an interval or collection of intervals, or indicate that it is the empty set:

   (a) (1 point) A^c

   $\boxed{[-1, 4)}$

   (b) (2 points) $A \cap B \cap C$

   $\boxed{\emptyset}$

   (c) (2 points) $(B \cap A^c) \cup C$

   $\boxed{\{-3, -1, 1, 2, 3\}}$

   (d) (2 points) $C \setminus B$

   $\boxed{\{-3, -1, 1\}}$

   Find the following if they exist or indicate they do not exist:

   (e) (1 point) $\min(A^c)$

   $\boxed{-1}$

   (f) (1 point) $\max(A^c)$

   $\boxed{\text{DNE}}$

   (g) (1 point) $\inf(A^c)$

   $\boxed{-1}$

   (h) (1 point) $\sup(A^c)$

   $\boxed{4}$

2. (10 points) Determine if each of the following statements is TRUE or FALSE by completely filling in the corresponding bubble. In either case you should provide a short explanation or sketch of a proof. **You do not need to provide a formal proof.**

(a) For all $x \in \mathbb{R}$, if $x^2 \geq 1$ then $x \geq 1$.

○ TRUE

● FALSE

counterexample:

If $x = -3$, $9 \geq 1$ but $-3 < 1$

(b) For all $x \in \mathbb{Z}$ and for all $y \in \mathbb{R}$, $y \neq 0$, $xy < 1$.

○ TRUE

● FALSE

counterexample.

If $x = 2$, $y = 1$ $\quad 2 \cdot 1 \geq 1$

(c) There exists $x \in \mathbb{Z}$ and there exists $y \in \mathbb{R}$, $y \neq 0$, such that $xy < 1$.

● TRUE

○ FALSE

example.

$x = -2$, $y = 1$ $\quad -2 \cdot 1 < 1$

(d) For all $x \in \mathbb{Z}$ there exists $y \in \mathbb{R}$, $y \neq 0$, such that $xy < 1$.

● TRUE

○ FALSE

- If $x < 1$, $y = 1$.
- If $x \geq 1$, let $y = \frac{1}{x+1}$, $\frac{x}{x+1} < 1$

(e) There exists $x \in \mathbb{Z}$, $x \neq 0$, such that for all $y \in \mathbb{R}$, $xy < 1$.

○ TRUE

● FALSE

If such an x existed, set $y = \frac{1}{x}$ then $x \frac{1}{x} = 1 \not< 1$.

3. Consider the following list of statements:

   1. For all $x \in \mathbb{R}$, if $0 < x < 1$, then $\frac{1}{x(1-x)} \geq 4$.
   2. For all $x, y \in \mathbb{R}$, $|x| - |y| \leq |x - y|$
   3. Prove that the sum of the squares of the first n natural numbers is $\frac{n(n+1)(2n+1)}{6}$.
   4. For all $x \in \mathbb{R}$, either $x + \sqrt{3}$ or $-x + \sqrt{3}$ is irrational.
   5. Let $a_1, a_2, \ldots, a_n \in \mathbb{R}$. If $a_1 a_2 \cdots a_n = 0$, then for all $n \in \mathbb{N}$, $a_i = 0$ for some i with $1 \leq i \leq n$.
   6. For all $x \in \mathbb{Z}$, $x(x+1)$ is even.

   (a) (2 points) Write in the box provided the numbers corresponding to two of the statements above where proof by induction is the best choice of proof method.

   $\boxed{3, 5}$

   (b) (2 points) Write in the box provided the numbers corresponding to two of the statements above where proof by contradiction is the best choice of proof method.

   $\boxed{1, 4}$

   (c) For each of the two numbers you wrote down in part (b), write down what you would assume to start your proof by contradiction.

   (a) (2 points) First:

   Assume for all $x \in \mathbb{R}$, $0 < x < 1$ and $\frac{1}{x(1-x)} < 4$.

   (b) (2 points) Second:

   Assume for all $x \in \mathbb{R}$ $x + \sqrt{3}$ and $-x + \sqrt{3}$ are rational.

4. (7 points) Use a **proof by contradiction** to prove the following statement. If you do any scratch work on this page you should clearly indicate what should be graded as your final proof. Your proof should be clearly laid out and your logic should be easy to follow. You do not need to use complete sentences and you may use mathematical shorthand.

   Let x be an irrational number and $y \neq 0$ be a rational number. Prove that the number $x \cdot y$ is irrational.

Proof: Let x be an irrational number and $y \neq 0$ a rational number. Assume, by way of contradiction, that $x \cdot y$ is rational. That is, there exist $m, n \in \mathbb{Z}$, $n \neq 0$, s.t.

$$x \cdot y = \frac{m}{n}.$$

Since y is rational, $y \neq 0$, there exist $p, q \in \mathbb{Z}$, $p, q \neq 0$, s.t. $y = \frac{p}{q}$. Thus

$$x \cdot y = \frac{m}{n} \Rightarrow x \cdot \frac{p}{q} = \frac{m}{n} \Rightarrow x = \frac{q \cdot m}{p \cdot n}$$

by the closure of the integers under multiplication and the fact that $p, n \neq 0$, we have $q \cdot m \in \mathbb{Z}$, $p \cdot n \in \mathbb{Z}$ $p \cdot n \neq 0$ so x satisfies the definition of a rational number. This contradicts the fact that x is irrational. Therefore our assumption that $x \cdot y$ is rational was false and $x \cdot y$ is therefore irrational. $\square$

5. (7 points) Use a **proof by induction** to prove the following statement. If you do any scratch work on this pag e you should clearly indicate what should be graded as your final proof. Your proof should be clearly laid out and your logic should be easy to follow. You do not need to use complete sentences and you may use mathematical shorthand.

For all $n \in \mathbb{N}$

$$2 + 4 + 6 + \cdots + 2n = n^2 + n.$$

Let P_n be the open sentence

"$2+4+6+\cdots+2n = n^2+n$"

Base case: $n=1$. $2 \cdot 1 = 2 = 1^2+1$ ✓

Inductive Step. Assume P_n holds for some n.

Then prove P_{n+1} is true.

$$2+4+6+\cdots+2n+2(n+1) = n^2+n+2(n+1)$$

$$= n^2+n+2n+2$$

$$= n^2+2n+1+n+1$$

$$= (n+1)^2+n+1. \quad \square$$

6. (7 points) Prove, using the **precise definition for the limit of a sequence**, that

$$\lim \frac{3n+1}{4n-1} = \frac{3}{4}.$$

Please write your complete proof on the bottom half of the page labeled "**Proof.**", only work that appears there will be graded. Your proof should be clearly laid out and your logic should be easy to follow. You do not need to use complete sentences and you may use mathematical shorthand. You may use the top half of the page labeled "*Discussion.*" for any scratch work. I may look at the scratch work if your proof is incorrect to help maximize the partial credit given.

Discussion.

Want $\left|\frac{3n+1}{4n-1} - \frac{3}{4}\right| < \varepsilon$

$\Rightarrow \left|\frac{7}{4(4n-1)}\right| < \varepsilon \Rightarrow \frac{7}{4\varepsilon} < 4n-1$

always (+).

$\Rightarrow \frac{7}{4\varepsilon} + 1 < 4n \Rightarrow \frac{7}{16\varepsilon} + \frac{1}{4} < n$

Proof.

Let $\varepsilon > 0$ and let $N = \frac{7}{16\varepsilon} + \frac{1}{4}$. Then $n > N$ implies $n > \frac{7}{16\varepsilon} + \frac{1}{4}$, hence $4n > \frac{7}{4\varepsilon} + 1$, hence $4n - 1 > \frac{7}{4\varepsilon}$, hence $\frac{7}{4(4n-1)} < \varepsilon$, and hence $\left|\frac{3n+1}{4n-1} - \frac{3}{4}\right| < \varepsilon$. This proves $\lim \frac{3n+1}{4n-1} = \frac{3}{4}$